

ActInf Textbook Group ~ Cohort 1 ~ Meeting 13 (Feedback, Next steps, Projects)

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Hello everyone it is meeting 13

of the active inference textbook group cohort one it is the final meeting of this section

it's July 28th 2022 and we are going to be especially thinking about feedback and structuring

on this textbook group on how the first half of the first cohort went and also ways that over

quite literally the coming minutes and hours and days we will be able to implement structures

to improve our experience for those of us who are continuing on to part two cohort one

as well as those who are joining for a new cohort in the onboarding page there are two columns

with check boxes and clicking these check boxes will sign you up to continue on in chapter six

through ten for cohort one as well as or alternatively to join in chapters one through five with cohort two

so select one or both of these boxes if you'd like to be continuing on

that's the onboarding note and again other than that which i hope those who are here or watching live

update according to how they want um other than that we're going to be talking about future textbook

groups and other things like that and then also just on the informational briefly on the future textbook

group page here's the link to share with people to join future textbook cohorts so to join in and then

also please fill out this feedback form it's really important that even if people think they have limited

or non-valuable feedback that they provide it because whether it's on this page directly where everyone can

edit and so every thought that somebody has like even if right now they can just edit this page or in

this more anonymous and asynchronous mode or through direct communication but we're experiencing it and

so we have the best insight at this point into how to improve it and the affordances that we're seeing

today or the the state updates or however you want to use ontology terms to describe it but what we're

seeing today we can do so we should have a fun discussion today on some of those topics

and then also look towards project ideas many of which are related to the educational process

and um others which are building on education into research and application and so there will be spaces for

those as well let's begin though on this future textbook groups page and anywhere where people want to jump in

writing or speaking

how are we going to

summarize and distill the experience of the last few months in this textbook group

how are we going to do that in this book and bring it towards improvements in our coming cohorts

you

from

Yeah, I think having worked examples, as I've talked about and discussed with Brock, like in Jupyter notebooks, or even simple, like did you understand the material, like question and answer, like quizzes at the end of each chapter, or something like that, I think would just be useful to gauge your own understanding, like if there is a definition or something that we can directly refer back to in the text, that'd be useful.

Okay, worked examples.

Yes.

Great.

And the questions could also include little tests of knowledge.

Not every question needs to be throwing a baseball into the deep unknown.

It can be literally clarifying for someone who believes that an answer is in the known unknown space.

I believe there's an answer for this.

I just want to clarify for my understanding.

But also, once you've clarified that, and especially if you did just clarify it, then write that example.

Because many of these are opportunities for worked examples.
examples.

And worked examples work in certain situations.

For formalisms, for example, it might also work for formalisms that aren't mathematical.

Like, here's the system this way, or this phenomena.

How do these ontology terms map?

Or if this is the blank, then what is the other thing?

Okay, so we have...

Okay, anyone else?

Just go for it.

Or let's just continue to edit and organize and add notes and just have more.

Because many people are having many thoughts on directions to go.

And just like, what is the space of some improvements that we can explore?

And then how will we be implementing some of these changes?

And then in cohort two, when someone's like, it could be this way, and they thought of something that's new and not on this list, we add it.

If it's on this list, we say, awesome.

We agree.

Join us and do it.

Please do it.

Please do it.

Please do it.

Please do it.

Please do it.

Please do it.

Please do it.

Please do it.

Please do it.

Thank you.

Anyone else with raised hands or we can look more at a few other sections before going to the projects.

Thank you.

Thank you.

When I see different regimes of attention, I think one way that we can structure that is by clarifying what practices we're explicating and where those practices can take place.

And it's like an infinite game to structure beneficial engagement.

And where the rubber hits the road is like being clear about what practices people can engage in.

So if we have math, like not everything has to be fractured out into sub sub sub groups.

Just including many of the practices.

Like if you want to do art about active inference.

Here's a practice.

And then if people are engaging in that practice, it does like create kind of like a group or a guild engaging in that performance.

So however people think like what are useful minutes and seconds.

That actually help learning active inference.

And we'll be able to have a perspective on that question in years from now.

That's very different.

Based upon better understandings of competencies and the fundamentals and the approach to the area.

However, I think the practices table and providing feedback and then.

Sharing and communicating how we're using practices to increase understanding could be very important for those who are learning and learning how to learn.

I think to build up on sort of like blue suggestion.

I'm thinking that.

Like in some of the courses that I've taken.

You have like the other participants giving.

So like feedback.

As part of the learning process, you give feedback on like the other participants work.

And so it could be that it's like there's an exercise and everybody has to like fill it up.

And et cetera.

And then the other participants.

And so it could be that it's like there's an exercise and everybody has to like fill it up.

And et cetera.

And then the other participants or like it could be like two or three.

And so not everybody has to do everything.

So it's like, yeah.

To the people you read like, it's like Jessica's like answer and give a feedback and you know, so on.

Because in that way by having to critique or like analyze all the people's work.

It helps you think through the material itself.

And yeah.

So that's like another way of also like learning.

Again, there's also missing like, yeah, you can make, you could be probably like making a mistake also not applying it correctly.

But it's like a tool to help you think through the material by analyzing somebody else's work.

While also, you know, initially also having to do it yourself and then hearing also the feedback that other people are going to be giving you.

So I was kind of like an idea that came to mind, like making reference from like other experiences.

Thank you, Jessica.

Yes.

I think that's a great area to keep in mind.

And to balance.

As our cohorts build and so on, like who's on the supply and who's on the demand side of that.

So to speak, like who is actually offering the feedback.

And then who is the people who are wanting feedback.

Like some of these projects are things where the person.

Was able to be.

Brought in to another project.

Like.

But like Noah's and Mike's work.

Bringing them into synchronous and asynchronous resources that are already happening.

Provided them with at least a first step.

Or continuing on their journey.

Others are.

They're in different areas.

And so how to keep it open when people have ideas ranging from educational improvements.

To robotic deployments.

What kind of feedback they need.

And then.

How to balance like the excitement around applying active inference.

With.

What it means to learn.

Like there are biochemistry classes.

It's not all just biochemistry lab.

But who's to say that's even.

The best way.

Or anything like that.

It's just.

We want to make sure that.

We.

Keep it all in mind.

Jessica.

Um.

Yeah.

And I think this.

Suggestion idea.

It's.

It's not connected to.

Someone's project.

Or in a 16 project.

It's more like.

An exercise.

So there's like.

You know.

Short exercise.

That it's going to be testing.

You know.

Variational inference.

Or something.

Like that.

And.

And then.

Yeah.

Like every participant in the class.

Go.

Yeah.

Actually does the exercise.

And then it's assigned.

To.

Other people in.

The cohort.

And to.

Analyze that person's.

Submission.

So it's like.

Yeah.

Three people in the course.

Like analyze.

Like analyze.

Like.

Like.

Just because.

You know.

Submission.

And give feedback.

And like.

Yeah.

It's going to be like.

For everybody.

Everyone.

And we'll be getting.

Three.

Three.

Three feedback.

On like.

Like.

How they applied the.

Terms.

Or like.

You know.

There are other ways.

That they could be done.

Etc.

And.

I mean.

If some of the things.

Are like.

Math related.

Like.

I will have to see.

And.

Cause.

Then the answer is going to be more clear.

It's not going to be as subjective.

Um.

But it's more like.

You know.

There's a short exercise.

Everybody does the exercise.

And then some people.

Then get to review that.

Um.

You know.

That answer of the person.

And give feedback on that.

So it's like.

Separate from.

An actual project.

Okay.

Makes sense.

It's kind of like a combination.

Of.

Having a good example.

Questions.

With an organizational structure.

And then.

Like you said.

Balancing.

Like where.

There might be.

A clearer.

A clearer.

Uh.

Measuring stick.

Like.

Was this.

The right.

Equation.

Or the right calculation.

Or not.

Versus.

How did.

This usage of affordances.

In a new context.

Make sense to you.

And.

Um.

Also.

It's not just about.

Getting.

Feedback on the person's idea.

Like.

The practice of.

Of providing.

Constructive.

Ideally.

Written.

Comments.

Is super.

Important.

The secret side.

Of the questions.

Is.

The discourse.

Um.

Where.

The kinds of things.

That.

If.

If.

If.

If.

If.

If.

If.

where the kinds of things that
questions and ideas that
send people on thought excursions
and research rabbit holes
we can
collaborate on that response
together and it doesn't have to be the
end all be all
but sometimes by putting
like a seed into the workspace by putting
like a prompt or a question
10 people could write for
5 minutes and then there would be 50 minutes
of writing on that topic from different perspectives
whereas
if there's 10 people also there could be 5 minutes of discussion
which might add more seeds onto the table
and it might
continue to invoke
whimsy

or it might continue to instruct
if there was a broad uncertainty that the 5 minutes of discourse resolved
yeah
yeah
yeah
yeah
yeah no i was gonna say like
i think like
for the questions
and it's like
somebody has like a question that they're trying to understand
and everybody's giving it the crowd source
answer and helping you know sense make
on the meaning of
on how things are
with the other exercise
i think similar to i think what blue wants to do
with like the short
um
like exercises to test whether people know the concept or not
and that
it's just like adding like
by having people critique
all the people's work
and having to like critically think
through the answer
is another level of learning
because you it's not just like oh i think this is the answer to this
it's like no i actually have to understand
and one the concept that i'm using to evaluate uh an answer
and then also evaluate that person's answer
and kind of compare and contrast
and that like having to make that exercise of critical thinking
i think helps people um learn like in a more in-depth way
because you have to think through your own thinking process
and like what things mean what somebody else was saying etc
um and yeah and i i've seen it applied to like writing i mean i guess it's maybe like
easier to like writing courses and things like that um but yeah it's like another way of um

yeah like processing through the information

and and then also um you know so maybe like reduces the burden

or maybe like a person having to like go through everybody's answer

and making sure like yeah like i am the teacher correcting everybody's answers

like it's yeah you're sharing the responsibility around everybody in the group

of you know like kind of answering you know like you know evaluating people's like responses to an exercise

thank you brock and then anyone else with the race hand

uh just to agree with jessica and the last part would i was going to add that but um

yeah but just systematically created creating a systematic affordance for people to review or provide some validation that it's correct or on the right path or whatever um will reduce the burden there and uh teaching

teaching it's probably the best way to learn um

um

it's

uh

learning how to learn is it's like difficult to do without um teaching

learning

like learning

self-learning is about teaching yourself

you get better at teaching you get better at teaching yourself

thanks um

which

oh continue

just the

circle back that does kind of require some meta structure to like

um

or source them or whatever

and then provide them

um

so i'm not sure

outside of the

project that i put on there

and

maybe

it's kind of related to jakub's as well but

outside of those two things i'm not sure we have a clear

affordance to do that yet

so

for
coming up with examples of
yeah like examples to work or whatever yeah
like sort either sourcing them and then putting them in a structure that says
oh this is for
section the chapter one
or whatever 1.1 or whatever
this is a
so not questions that are related to specific parts of the textbook
with people to work through but what kind of table or structure
could be
i guess virtually the same
the same but instead of a question
a question somebody is asking
it's an example
for other people to answer
maybe with an answer
that they can see like
if they want to reveal it or whatever
but i'm
you these
does that make sense
yes
yes
um
that's that's what i think
that's a

it doesn't have to be in this exact
but just make sure that we have written it down
and everyone can be
contributing to like write it down and what the structure is because it's not like this is a
impossible affordance
we can spin up the table right now and template it and just start to design it and over the coming weeks we
have time to digest all these different inputs to make it so you know in a few weeks it will fly by but we might
be able to be implementing these things
i wanted to ask what practices did people feel like they engaged in fruitfully whether it's on here or not just
when you felt like you were participating in the active textbook group what did you feel like you were doing

and how did it help equation translation that was like the most helpful thing that you were doing and how did it help

equation translation that was like the most helpful thing that i did

um and i don't think it's there in the well i guess it's kind of there it was like to copy the equations down but like putting them into natural language was probably like the thing that i did that was maybe most useful for myself and others

um yep it was a great practice and um it it leads to and scaffolds so many other why is it this way what does this mean what are the examples it's not like the prerequisite per se it's possible to ask questions without having the natural language description

description but whether it's a basic or a multi-line equation

it helps bring consilience to that discussion because there's a non-zero probability that then a term is going to be used loosely or it's going to be used one way and then it's going to be conversationally used slightly differently or people will not agree on which ontology terms are invoked or not

which can then have every equation start to spiral off into every possible ontology term and they very well might actually connect that way but being able to see what is variational free energy and if there's some term that is being connected to variational free energy that's not being described in the equation that's fine

that's fine like least squared error isn't in the linear regression y equals mx plus b so there can be formal and informal concepts that are super important for understanding that aren't with a symbol but this was a great way to be clear what the symbols actually are

what other practices did people feel like they benefited from

what would they recommend for a daily schedule for somebody who wanted to learn a lot

i think uh plus one

I think plus one there on Blue's writing the equations, which is maybe a super simple but open-ended almost kind of version of Jessica's suggestion.

But also in general, just adding stuff to the coda, asking questions.

Like, if you read and you don't ask a question, did you really grapple with it at all?

I don't know.

How do we improve thinking by writing and the actual number of typed words that people add in as a proxy?

During a discussion is one thing, but even just afterwards or on their own time frame, how can we have 10 people who do read a chapter, just scaling numbers?

Each of them could think of several questions, ranging from simple, I wasn't sure about, to simple, again, like more like an example one.

Like, what are the three kinds of this?

And it's totally known and the answer is fully listed to research your questions.

But how could we have many more inputs in terms of the questions that people are asking?

Do people feel like they wrote questions in their own notes, but didn't bring it into shared workspace?

Or do people feel like for themselves or others that they observed that the questions weren't

actually phrased?

Jessica, then anyone else?

Yeah.

I was going to share more about the student practices, but I guess, like, first of all, like the last question you mentioned, there were things that I didn't put in the general textbook thing, which were more about parsing information, which was like helping me connect the dots and synthesize information, which Ali actually helped me a lot also with some of that.

But those things were sort of like personal for me to like try to make sense of the material. In terms of the student practices, I like initially I was thinking to do like every day, you know, like an hour each for like the reading and maybe like an hour for the math to kind of get foundational things.

But then I found it was like better for me to maybe dedicate like a longer period of time on one day or two days for the reading.

So I suppose to like a little bit of the reading or like chunk more on, you know, two, three hours in one day and then maybe like follow up another day with like a longer period of reading as well instead of daily reading a little bit.

And I think that's more because I guess for memory, I mean, they do say to do like repetition for memory, but I think like for me it was like, well, read longer and, you know, have like the material fresh in your mind as like reading as opposed to going every day.

And then like for the math, because I guess there's less of the, you know, keeping the material fresh in your mind, like, yeah, like daily I was trying to do that.

So then it fell off a little bit on that part.

But yeah, like kind of at least one every day, like four days a week, at least like an hour

or so, like trying to focus on the math.

But yeah, that's kind of how my practice, you know, yeah, got, you know, developed over time through the course.

And that sounds like a weirdo with a weirdo with a weirdo with a weirdo with a weirdo with weirdo with weirdo with weirdo with weirdo with weirdo with weirdo with weirdo with

so the textbook is for as the

like the research people who want to apply it in their research

if we remember like from the introduction provide readers with the

ability to understand and use active inference in their own research

we'll have a better sense after part two with a more applied side

who might the next cohort of the textbook group

be in and communicate to

yes over many iterations and you know it can go many ways but when people are thinking like

in the next few weeks

who

blue row okay

okay

can we review which questions and understand

because i i don't quite understand how this could

like so if we go to the um like if you want to check out the questions check out like chapter

two like and a lot of the questions were so if you scroll down um keep scrolling keep scrolling

like um i'm trying to think about

which ones that were skipped let's see

and also there's no skipping there's what we can address in the one hour or two hours

right but i mean like we can raise our attention to all the important questions

yeah it was just like um and then i realized like most of my questions were like of a largely

technical nature so let me scroll through the coda um and see if i can find them but it was like

in chapter two it was like let's see

like the general thing to me is how how can we um among other things prevent question self-censorship and filtering

well so for me that question isn't just like this one like genius meets genius perfectly framed

specific answer that will be understood in one sentence but that question is like a broader space

of prompts that don't necessarily have any prior on whether it has to do with an application or it

could be a a technical question well so it was reference yeah good it was early it was like in

chapter two and chapter three and there were like a lot of questions um i'm trying to like look at the

specific i'll have to look at my own coda instead of watching you beautiful people but um it was and like it just happened early like in those meetings um and and it's not that there wasn't time like that we didn't address the questions in order it was like let's skip this because it's technical let's skip this because it's technical let's skip this because it's technical so i just stopped putting my questions in because i feel like they're technical like what does this little squiggly symbol mean and and that's why like for me the translation of equations into um natural language was the most useful thing because i feel like all of my my questions are technical like related to the math so i stopped entering them into the coda because they were just skipped over okay thanks well yeah i appreciate it um again i personally hope that we can have structures that uncouple the perceived importance of real-time speech from the contributions that people make so that we can be really clear about like which questions we're going to be addressing in conversation but and and use conversation for what it can bring many things and then have 200 questions in chapter two with many up voters or with whatever mechanism that brings different questions to the fore so there isn't even like the plausibility of being skipped it's just a question of like what will we raise up and focus on individually and as a group from the many many questions of different kinds that support a beginner with fundamentals and a pre-beginner with background and somebody who's asking research questions as well so i think one simple thing is uh if if you go back to the questions um table like i think you have a um interest column here um interesting yes even better um maybe there is a uh what we will address in the net coming live stream or a session kind of table that is taken from the questions table but the questions table has no such like projection of what is interesting or not in some especially in some one-dimensional kind of like yes or no thing um so it's just a separation of random things prompts questions and then there is some process to get it to what will what will be addressed in the live stream that um again there's ways to like try to do things like randomization or trying to like ensure that you know we don't continuously ask focus from one person and or things like that the popularity doesn't kind of win the day um there's more complex ways to do this that are kind of almost like decision making processes like um ephemeral group processes and things like that but um um yeah i think just separating them it would be helpful there at least um another thing might be like zoom you could do breakout rooms maybe that's not the format but um again just uh that's kind of an ephemeral group processing like like perhaps everyone doesn't want to focus on the same questions should there be temporary temporary split of merging of attention and that for some short period of time and then recombine don't know what that means for a live stream or recording precisely but yeah that's possible solution yeah thanks gather is awesome for these things but it does require clarity around how it's going to be done like mass one-on-one breakouts mass smaller great breakouts

and rearranging people using their spatial location to express different things and
assemble into different ephemeral teams
okay let's look at some
project ideas okay so for those who are here or not i hope i'm not misrepresenting anything
um mike brought up some token economies that's something that we've been actively going towards
in active blockfronts
um
so i'll just add just a note column
inactive blockfronts
as well as as mentioned earlier um
working with noah on some more some related topics
okay audiobook recording of a textbook just as we go through these projects
um i've recorded the first chapter and the preface type materials
do people think that an audio recording
has
utility
and if so is anyone interested to pursue this avenue with blue and i because it's very defined
and it involves
primarily the equations having natural descriptions and then just reading
the
and it can go many other directions too but this is like um
kind of a clear accessibility affordance
for different participants that might be interesting as well as just get out there
on like youtube and other channels as having that
artifact blue
um i do think it's useful obviously uh if i've elected to to i didn't realize i signed up to be
the participant lead but i'm so happy to do it
co-lead co-lead um but but i do think it's useful as someone who drives like a considerable amount of
time like every day um if it's not like i'm not a very good auditory learner but it does help to like
reinforce the concepts um and i i just wanted to mention um even if like you're not comfortable
perhaps like doing the math or reading the textbook um daniel thinks he can read it until it's perfect
um i i would prefer to do some video or audio editing um so that's also an option is to do audio
editing like where he's like you know you occasionally like stutter daniel does it very infrequently
but i but i think there's the occasional tripping over a word or like um saying a word twice like
when you lose your place or something like that so video or audio editing um i do it with windows
video editor but like audio editing is also another possible affordance there and it just would involve
like you know reading along with the textbook to make sure that it's accurate that no sentences are
skipped and then to just splice out any like mishaps or just ask him to re-record the you know markov

blanket for precision just so you can place it over the trip trip up if needed yeah thanks definitely we would do a multi-part i i just did one pass on the first chapter no equations etc draft but okay anyways just so people who are um interested here or re-watching can know about this okay resources for conducting research or developing active inference agents ways to facilitate the starting of that work for participants just a seed idea um a lot of this is going to be explored in part two because it's a recipe for designing the models and we'll have a lot of infrastructure around that as well so i hope whoever wrote this continues in that interest and builds that starting point for conducting research or developing agents based around what they learn and see in part two okay um okay also typing just directly in the coda has is a little better than comments because sometimes it's hard to understand what the comments are um like referring to like okay complete mathematical overview of active inference derivations of key results maps of equations yes all awesome thank you yakup for suggesting this um i think the closest related works are in the act in fontology and in knowledge engineering and other areas too um and i hope that we're on this path so however yakup or anyone else wants to connect on that okay active foundations um so here we added this page um if um brock or anyone else wants to add any comments or describe how some of these comments can be integrated with textbook group um short answers i'm not entirely sure the precise way to integrate it with the textbook group but um the textbook group um they're just um perhaps this is related to questions making questions um removing the stigma or whatever like it's just helpful i think to have some affordance or regime of attention or whatever thing that says hey if you don't understand this thing about active inference that's a foundational thing like okay i get entropy but and i get energy like the basic concepts of them but like when you're subtracting them what the hell are we talking about you know um and then maybe there's you know some follow-on questions about mathematically what entropy is mathematically what is energy or you know what is shannon entropy specifically what is the difference these sorts of things um a it's that in affordance for that but b it's also just acknowledging like um a there's a wealth of knowledge already there's a wealth of knowledge already here it's not a complete mystery what those back background foundational things are it's or what learners might want to get out of um some relatively short course to like get up to speed and feel real confident reading um the textbook um and so there's a kind of top-down uh affordance there too i think that needs to be made for um people that can say hey there's a bunch of equations over or examples over here that are these are the most helpful ones i've found for understanding what bayesian inference really is um how that really works um so i guess it would be those two things but if you look into the project um notes there on that project idea um thing it's there's a little more uh

detail of um yeah just a couple bullet points there of just it's it's really hard for people
i think to get into a new thing if there's not a model if you do chemistry without Bohr's model like
good luck if you have to jump especially to something like electron probability clouds or
something it's just like why would you do that when you have a perfectly good periodic table
and simple discrete you know and it's enough even to do basic physics right like Newtonian stuff right
and so um you need a model like i think a lot of people history is helpful there very minimal amount
but just uh this is where this came from this is why it's relevant um it's really hard for people i think
to blindly calculate without that sort of simple context um
so i just kind of made a very rudimentary still kind of scaffolding it um
Coda it's already linked in that project row but i'll put it in that sub page there too um super
work in progress but um
yeah i mean the idea is just there's these questions are going to be asked over and over and over again
there will be people coming from a more mathematical more physics a more neuroscience a more from
different
backgrounds and so why pretend like that's not happening or reinvent the wheel every time somebody asks a
question like just having a central resource where people can get a lot of value but also maybe
ask a question that's a little more
they might not want to ask and seem like it's too simple or it's too whatever um in
the live stream so
okay thank you
any other comments or thoughts on this otherwise we'll just continue on for the final minutes
okay
textbook enrichment yes i hope everyone continues to annotate and figure out also things on their own and
explore and like we're building something really powerful with annotation so i hope that people can
continue to be enriching
with the structures that we're building um but yes so much of what this group has been about has been
about textbook enrichment and improvement
robot modeling unless anyone has other comments i think learning from JF and um staying in part two
will help and um yeah Jessica and then anyone else
uh
sorry i think i didn't raise my hand from yours
yes well any general comments that people want to provide on the textbook group
is
i think the live streams are super important
for the lab and the whole space for a maybe obvious reason the textbook group is equally important
i think these are like the two most important things that can add value to this space and the lab
participants so that's it indeed glad that those who have made it this far see that
and

over the right time scales in the right way

we see more and more vistas to increase the accessibility and rigor of active inference

formats so i hope we can hold that frame and focus our attention

to provide these functionalities

because if it's stateable it's doable

in a way

the right way

and if no one from the first cohort wants to be an active facilitator or steward

for textbook group two some of these changes will be integrated

the more people who want to step in to steward

in our participatory context are going to be leverage points for so many more people who come through

so there's like multiple levels of affordances for those who see them

especially at the end of this section

and um

it's been quite a few months

any last comments

okay i'm going to stop the recording in three seconds unless somebody wants to say something

okay